

Developing of an Geometry-aware AI model for Predicting Stress Distribution in Aluminum Wheels under Impact

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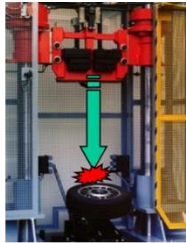
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ABSTRACT: This study proposes a geometry-aware Graph Neural Network (GNN) model for predicting stress distributions under dynamic impact on aluminum wheels. The model incorporates a weighted loss function and an edge reconstruction technique to improve accuracy in stress concentration regions. Using high-fidelity 3D finite element data from 85 wheel designs, we applied edge augmentation to enhance generalization with limited samples. The GNN model achieved a MAPE of 7.0% in critical regions, significantly reducing computation time compared to conventional FE analysis. This result demonstrates the feasibility of using GNNs for early-stage structural performance prediction in automotive development.

KEY WORDS: vehicle development, computer aided engineering, deep learning, stress prediction (B2)

1.INTRODUCTION

The durability and strength performance of vehicles are critical development factors that directly affect product quality and safety. Aluminum wheels, in particular, have a direct influence on vehicle stability through their contact with the road surface. Failure of an aluminum wheel during driving can pose a severe threat to driving safety. Therefore, it is of paramount importance to predict and design wheels with sufficient strength performance to withstand external impact loads.



(a) Wheel Impact Test

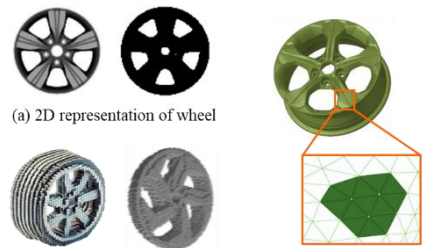


(b) Aluminum Wheel shape

Fig.1 Wheel 13degree Impact Test and Analysis

Currently, the strength performance of wheels is evaluated through a 13-degree impact strength test, which simulates the impact conditions experienced by wheels during vehicle operation (Fig. 1). In the wheel design process, strength prediction methods primarily rely on CAE (Computer-Aided Engineering) analysis based on finite element methods. However, this approach requires substantial time and computational cost, which limits its ability to accommodate the development of wheels with novel and diverse designs. To overcome these limitations, as in many other industrial domains, research on accelerating performance prediction through

the application of AI has been actively pursued in the areas of vehicle development and wheel performance prediction. In prior work, an AI model for predicting the mass of aluminum wheels was developed, thereby validating the feasibility of data-driven performance prediction.⁽¹⁾ However, the AI model trained on 2D images and structured design data failed to achieve the target performance in predicting disc stiffness and impact strength of wheels, primarily due to insufficient learning of 3D geometric information. Subsequent attempts utilizing 3D voxel data representation and automatically generated 3D data were made to address this issue; however, the prediction performance for stress distribution on the wheel surface remained inadequate due to limitations in detailed geometric representation.⁽²⁾ (Fig. 2)



(a) 2D representation of wheel
(b) 3D Voxel representation of wheel
(c) 3D Graph representation of wheel

Fig.2 Data Representation method for Wheel Geometry Learning

To resolve these challenges, a preliminary study was conducted applying a novel approach based on Graph Neural Network (GNN) technology.⁽³⁾ (Fig. 3). The proposed GNN framework achieved an R^2 of 0.965 for von Mises stress prediction on simplified wheel geometries, and demonstrated improved prediction performance and robust generalization capability through the application of a

novel mesh connectivity reconstruction technique. Although this study demonstrated the potential of GNN-based prediction models for 3D solid mechanics problems, it was still limited using research-scale datasets with relatively small data sizes and low complexity, making it insufficient for direct application in product development.

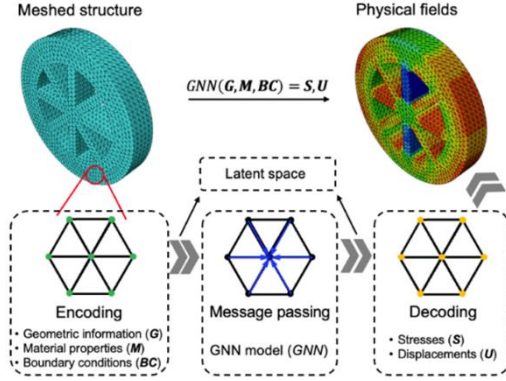


Fig.3 GNN Model Trained on Simplified Wheel Geometry Data

In the present study, the training data and analysis conditions were aligned with the actual development work standards to enable direct application in the real design process of aluminum wheels. Correspondingly, performance improvement algorithms were incorporated to optimize the prediction model. Through this approach, we aim to contribute to the early prevention of strength performance issues from the initial stages of vehicle design and to the enhancement of design quality. This paper provides a detailed description of the development process and performance improvement of the GNN-based AI model for predicting the impact strength performance of aluminum wheels and proposes a new paradigm for the design and performance evaluation of chassis systems.

2.Training Data

2.1.Graph Neural Network and Mesh Data

Graph data is a structure composed of nodes and edges that represent connectivity relationships between objects and is utilized for modeling complex relationships such as networks.⁽⁴⁾ Mesh models are similarly composed of nodes and edges, where nodes represent the vertices of polygons and edges define the boundaries between them, thereby representing 3D geometries (Fig. 4). Given the structural similarity between graph and mesh representations, mesh data can be converted into and expressed as graph data. Graph Neural Networks (GNNs) are models that take such graph-structured data as input and learn relational patterns. By constructing meshes in graph form, structural information can be effectively learned through GNNs.

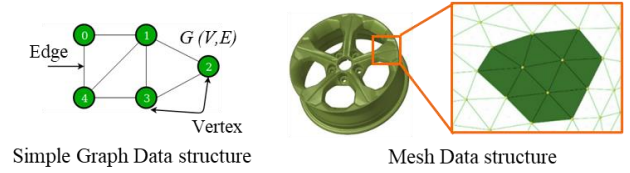


Fig.4 Comparison of Graph and Mesh Data Structures

2.2.Wheel Impact Analysis Data

From the perspective of geometric fidelity of the analysis model, the number of nodes in aluminum wheel analysis models following the current analysis work standards ranges from approximately over 200,000. Since the data used in this study employs analysis models of equivalent fidelity, the data size is up to 100 times larger than the 3,000–5,000 node-level data used in preliminary studies. Moreover, the data captures the full geometric detail of the wheel, enabling the pursuit of prediction performance on par with that of finite element analysis (Fig. 5).

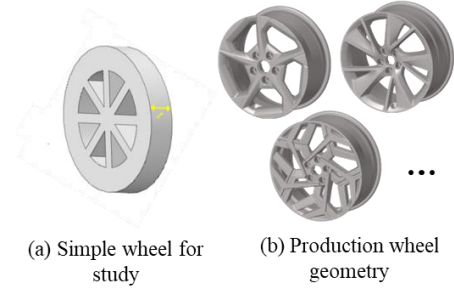


Fig.5 Comparison of Fidelity Levels of Training Data

Regarding the analysis conditions, the 13-degree impact strength analysis of wheels can be performed using either static or dynamic analysis methods. The static analysis method is suitable for initial research data construction purposes due to its relatively simple analysis conditions and shorter model setup and computation times. However, in this study, dynamic impact analysis results were used as training data, with the objective of applying the prediction model to actual development work. (Fig. 6)

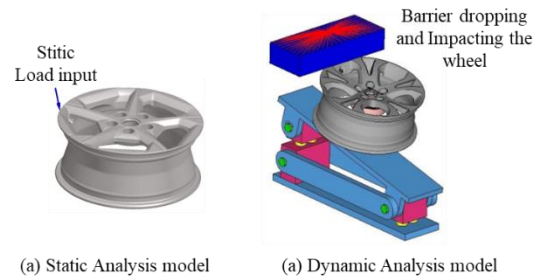


Fig.6 Comparison of Fidelity Levels of Training Data

That is, the training data used in this study reflects the practical analysis work standards, possessing geometric fidelity and analysis accuracy at the level of production-specification wheels (Table.1). However, as data size increases, the data dimensionality and training time also grow, necessitating appropriate data preprocessing and algorithm implementation.

Table.1 Comparison of Training Data

	Preliminary Study	Current
Analysis Method	Static Strength	Dynamic Strength
Analysis Conditions	Simple	Complex
Analysis Time	Small	Large
Wheel Geometry	Simplified structure	Production Geometry
Data Size (No. of Nodes)	3~5k	200~400k

2.3.Data Pre-Processing

The training data consists of analysis results obtained by evaluating the performance of the finalized design geometries of aluminum wheels using the dynamic strength analysis. For a total of 238 wheels, existing Abaqus analysis results performed under 2 to 4 different impact load directions per wheel were collected, yielding a total of 745 strength performance datasets. All data were automatically extracted from ODB files and PPT documents using Python and Abaqus scripts. The automation code was implemented to be reusable for preprocessing in future iterative data additions. The extracted data includes wheel design information, striker mass, and the stress and displacement data of all nodes at the time increment corresponding to the maximum stress. (Fig. 7)

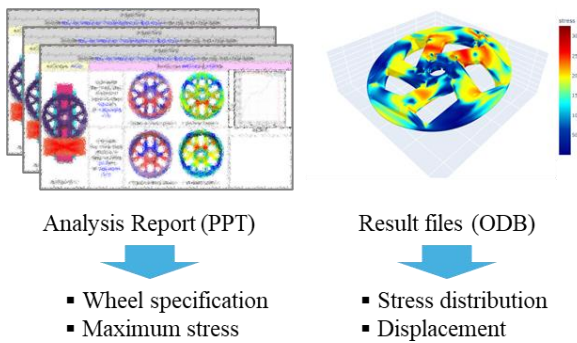


Fig.7 Automated Data Extraction

The mesh models used in the strength analysis are composed of quadratic elements. Quadratic elements include additional mid-nodes between primary nodes to enhance geometric representation accuracy; however, these mid-nodes do not contribute to GNN learning in graph data. Furthermore, the rim portion of the wheel

exhibits minimal geometric variation and has a limited influence on strength performance. Therefore, only the node data stored in the spoke region-the primary area of concern in strength analysis-was utilized, and this information was preprocessed into graph form for use as GNN model training data. By incorporating domain knowledge of wheel strength analysis in this manner, the training data size was effectively reduced to an average of 2.4% relative to the initial analysis result data. In this study, based on 290 Abaqus analysis results from 85 wheels out of the total dataset, the final training set of 580 data samples was constructed through approximately two-fold data augmentation using Random Edge Connection. The dataset was partitioned such that no identical wheel geometries appeared in both the training and test sets, and different wheel geometries were evenly distributed: 524 samples (77 wheels) for training and 56 samples (8 wheels) for testing (Fig.8). Although the number of data samples may be limited, recent studies have reported that GNNs can achieve high prediction accuracy with a small number of training samples for structures with complex geometries.⁽⁵⁾ This is attributed to the ability of GNNs to effectively learn and generalize physics-based local rules.

	#	Wheel Geometry	Impact load	Loading direction
Training data	1	A	P_A	θ_0
	2			θ_2
	3			θ_3
	4	B	P_B	θ_4
	5			θ_5
Test data	⋮	⋮	⋮	⋮
	578	Z	P_Z	θ_6
	579			θ_7
	580			θ_8

Fig.8 Training and Test data Configuration

3.Development of Prediction Model

3.1.GNN based model

In this study, a GNN architecture based on MeshGraphNets⁽⁶⁾ was employed to construct a model that predicts stress and deformation from mesh geometry, boundary conditions, and material properties as inputs. The output consists of six stress components and three deformation components per node. The model comprises a three-stage architecture: encoder, message passing, and decoder. The encoder maps geometry and boundary condition-based features into a latent space, the message passing module updates representations through information exchange between adjacent nodes, and the decoder restores the latent representations to physical quantities. Fig.9 presents a schematic

overview of the overall GNN architecture and the flow of each stage.

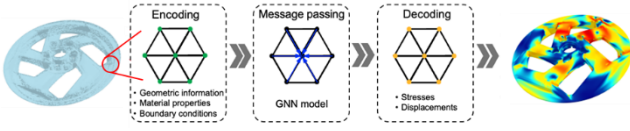


Fig.9 Schematic of the GNN model

3.2.Improvement of Prediction Performance

In wheel impact strength tests, failure predominantly occurs at localized regions such as spoke junctions or local notch geometries. In impact strength analysis as well, the stress concentration zones constitute only a small portion of the overall wheel. To ensure that the model assigns greater weight to such stress concentration regions during the learning process, a modified loss function was employed (Fig. 10). The application of the improved loss function resulted in a 10% enhancement in prediction performance.

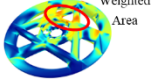
$$\text{New loss} = L_2(\text{prediction}_1, \text{ground truth}_1) + \gamma L_2(\text{prediction}_2, \text{ground truth}_2)$$


Fig.10 Modified Loss Function and Weighted Region

In addition to techniques applied during the training process, edge reconstruction and augmentation methods were applied to the graph structure of the training data. As mesh density increases in 3D structures, the message passing range becomes limited, making it difficult to capture long-range information. To address this, the preliminary study proposed a method of shortening message passing paths by constructing additional edges within a given fixed distance.⁽³⁾ In this study, building upon this approach, a geometry-aware edge augmentation technique was implemented. Rather than applying augmentation to all nodes, additional edges were generated by randomly selecting nodes from regions with distinct spatial characteristics: the boundary region (Impact zone) where the falling barrier directly imparts load to the wheel, and the central region (Non-Impact zone). (Fig.11) This enables critical information from the boundary region to be efficiently transmitted over long distances to other parts of the structure. During model training, the random seed for node selection and the selection ratio were set as hyperparameters. By generating multiple edge-augmented datasets using different seeds, the diversity of training data and the generalization capability and robustness of the model were enhanced. Furthermore, since edges are not generated for all nodes, the computational cost is reduced compared to the conventional method, resulting in shorter training

times. The edge reconstruction and data augmentation methods yielded an additional 3% improvement in prediction performance.

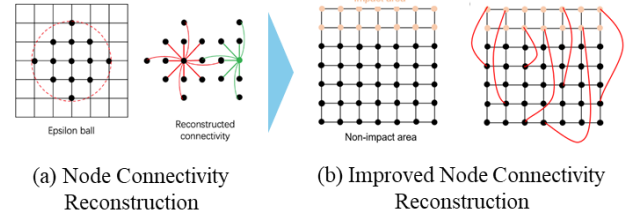


Fig.11 Improved Mesh Reconstruction Method

3.3.Final Prediction Performance

A comparison between the GNN predictions and finite element analysis results across various wheel geometries and impact strength conditions revealed that the GNN model with the weighted loss function achieved a MAPE of 7.0% for von Mises stress prediction in the region of interest. MAPE (Mean Absolute Percentage Error) is a metric that represents the mean absolute percentage error between predicted and actual values, and is well-suited for evaluating the prediction accuracy of von Mises stress, which serves as an important criterion for ductile material failure. In this study, by applying the weighted loss function and geometry-aware edge technique, the model's prediction accuracy was improved particularly in high-stress regions, enabling the prediction of potential failure locations. To accurately assess the prediction performance in the stress concentration zones of interest, the evaluation was conducted on nodes with stress values in the top 20th percentile (Fig. 12).

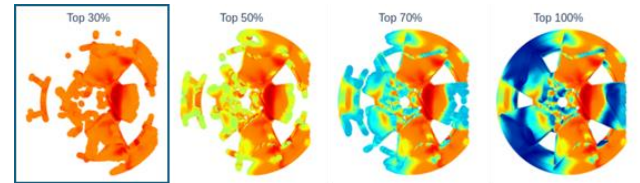


Fig.12 Comparison of Top N % Stress Regions

Additionally, the Kullback–Leibler Divergence (KLD) was computed to quantitatively evaluate the prediction performance of the stress distribution. KLD is a metric that measures the difference between the predicted and actual stress distributions, where values closer to zero indicate greater similarity between the two distributions.⁽⁷⁾

The test set data consisted of 56 samples obtained from 8 wheels (Fig. 13). That is, the trained GNN model demonstrated prediction performance of MAPE 7.0% and KLD 0.005 for the region of interest under impact strength conditions for wheel geometries unseen during training. It is anticipated that further

improvement in prediction performance can be achieved by additionally training on the remaining data not used in this study and by further narrowing the region of interest to specific stress concentration zones.

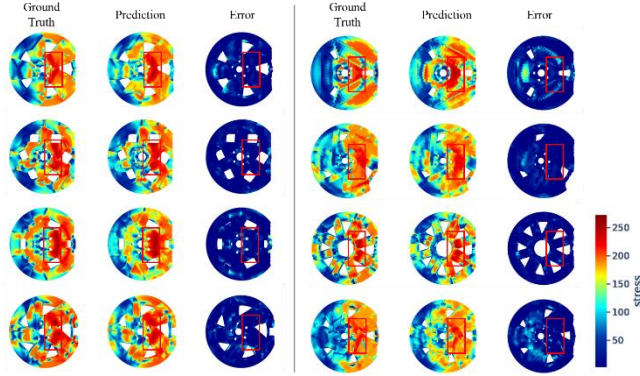


Fig.13 Impact Strength Stress Prediction Results of the Test Set

To analyze the influence of hyperparameters on GNN model performance, tuning experiments were conducted across various combinations, as summarized below (Table. 2). When the number of message passing layers and embedding dimensions were increased, improvements in both R^2 and MAPE metrics were observed by comparing Trial 1 with Trial 2, and Trial 3 with Trial 4. Variations in batch size did not significantly affect model performance, which is attributable to the limited number of samples in the present dataset. Regarding the optimizer, AdamW—an improved variant of Adam with enhanced regularization and generalization performance—yielded superior results, consistent with its known effectiveness in large-scale model training. Furthermore, a comparison between Trial 1, which employed the edge augmentation method, and the baseline results confirmed approximately 3% improvement in MAPE, thereby demonstrating the effectiveness of the proposed edge augmentation technique. Comparing the edge/data augmentation experimental results with those of edge augmentation experiments 1 and 2 in the table, it was found that cumulatively training with multiple batches of randomly generated edge-augmented data enables effective expansion of the training dataset, achieving the best performance across both the top 20% MAPE and KL divergence metrics. Through these hyperparameter tuning experiments, a balanced trade-off between convergence stability and prediction accuracy of the model was achieved.

Table.2 Hyperparameter Tuning of GNN model

	N / H / B	R2	MAPE	MAPE Top 20%	KL divergence
Baseline	15 / 64 / 2	0.700 ± 0.001	0.338 ± 0.038	0.085 ± 0.001	0.105 ± 0.009
Trial 1	20 / 64 / 2	0.698 ± 0.004	0.285 ± 0.042	0.076 ± 0.013	0.078 ± 0.010
Trial 2	10 / 64 / 2	0.696 ± 0.002	0.341 ± 0.018	0.082 ± 0.005	0.082 ± 0.015
Trial 3	15 / 128 / 2	0.702 ± 0.004	0.329 ± 0.071	0.072 ± 0.008	0.078 ± 0.020
Trial 4	15 / 32 / 2	0.685 ± 0.001	0.368 ± 0.048	0.085 ± 0.004	0.122 ± 0.034
Trial 5	15 / 64 / 4	0.684 ± 0.003	0.316 ± 0.027	0.076 ± 0.002	0.110 ± 0.011
Edge Augmented 1	15 / 64 / 2	0.691 ± 0.004	0.313 ± 0.045	0.078 ± 0.002	0.078 ± 0.019
Edge Augmented 2	20 / 64 / 2	0.693 ± 0.002	0.291 ± 0.021	0.077 ± 0.001	0.072 ± 0.015
Edge and Data Augmented 증강	20 / 64 / 2	0.712 ± 0.001	0.270 ± 0.025	0.070 ± 0.004	0.050 ± 0.009
N(Number of layer), H(Hidden dimension), B(Batch size)					
	N / H / B	Optimizer	Learning rate	R2	MAPE
Trial 6	15 / 64 / 2	Adam	0.001	0.676 ± 0.010	0.354 ± 0.026
Trial 7	15 / 64 / 2	SGD	0.001	0.660 ± 0.001	0.330 ± 0.002
Trial 8	15 / 64 / 2	AdamW	0.0001	0.667 ± 0.001	0.353 ± 0.045
Trial 9	15 / 64 / 2	AdamW	0.01	0.674 ± 0.001	0.302 ± 0.033

4.CONCLUSION

This study demonstrated that the proposed Geometry-aware GNN model can reliably predict the stress distribution induced by dynamic impact on aluminum wheels. This method offers the advantage of accurately identifying critical stress concentration regions while dramatically reducing computation time compared to finite element analysis, all while preserving high geometric fidelity. To achieve this, a novel loss function that assigns weights to stress concentration regions and edge reconstruction and augmentation methods were designed and applied. Through these techniques, large-scale production-level precision 3D data could be effectively processed. This study demonstrated that the GNN model can be effectively utilized for performance prediction at the early design stage based on actual vehicle development data.

The data acquired during this research could not all be used for training due to limited GPU resources. Furthermore, data from future production wheels subjected to analysis review can also be utilized for training. It is planned to sequentially train on such data to improve prediction performance. In addition, there are plans to apply the same framework to predict other performance aspects of wheels as well as the performance of other chassis components such as suspension arms.

The current study focused on the stress state at the time of maximum load; however, future extension to a time-series-based GNN model capable of tracking stress evolution throughout the entire impact event is feasible. Beyond this, research on improving GNN prediction reliability by leveraging experimental data is also possible.

As the computational complexity of dynamic analysis increases, GNN-based performance prediction models are emerging as a promising alternative. In this study, a GNN model capable of efficiently predicting the impact strength performance

of aluminum wheels was developed, and it was demonstrated that this model can contribute to accelerating the actual vehicle design and verification process as a data-driven model capable of efficiently predicting complex physical phenomena. Through future improvements in learning techniques, model architecture, and generalization strategies, the applicability of GNN models is expected to expand further.

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